



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

GENERATIVE AI LABORATORY PRACTICAL RECORD

Prompt Engineering — Zero-shot, One-shot & Few-shot Prompting

Submitted by:

C.PRAVEEN KUMAR

192472034

CSE-AI

Course Code: CSA6502

Course Name: Generative AI

Academic Year: [2024–2028]

EXPERIMENT 1: Zero-shot, One-shot, and Few-shot Prompting for Product Description Generation (Smart Fitness Watch)

1. Aim

To design and implement Zero-shot, One-shot, and Few-shot prompts using a Generative AI tool to generate a product description for a Smart Fitness Watch, and to compare the quality and consistency of the outputs produced by each prompting technique.

2. Objective

- To understand the working of Zero-shot prompting.
- To understand the working of One-shot prompting.
- To understand the working of Few-shot prompting.
- To analyze how the presence of examples in a prompt influences the structure, tone, and feature coverage of an AI-generated product description.

3. Prompting Concepts Used

- **Zero-shot prompting:** The model is given only a direct instruction, with no example of the expected output. It relies entirely on its pre-trained knowledge to generate a response.
- **One-shot prompting:** The model is given exactly one example of the desired input-output pattern before being asked to perform the actual task, which helps guide its style and structure.
- **Few-shot prompting:** The model is given two or more examples before the actual task. Multiple examples reinforce the expected pattern, tone, and format, generally producing more consistent output.

4. Input / Task Description

Product: Smart Fitness Watch. The generated product description is expected to include the following elements:

- Product name
- Key features
- Health tracking capability
- Fitness tracking capability
- Battery life
- Connectivity
- User benefits
- Attractive and professional marketing language

5. Zero-Shot Prompt

Write a professional and engaging product description for a Smart Fitness Watch, highlighting its health tracking, fitness tracking, battery life, and connectivity features.

6. Zero-Shot Generated Output

Introducing the PulseFit Smart Watch – your all-in-one companion for a healthier, more active lifestyle. Featuring continuous heart rate monitoring and SpO2 tracking, it keeps you informed about your vital health metrics around the clock. Track your daily steps, calories burned, and choose from multiple workout modes designed for running, cycling, and yoga. With sleep tracking, you gain insights into your rest patterns to help you recover better each night. The watch offers a long-lasting battery that runs up to 7 days on a single charge, and seamless Bluetooth connectivity syncs effortlessly with your smartphone to keep notifications, calls, and music at your wrist. Stay fit, stay connected, stay ahead – with the PulseFit Smart Watch.

[Insert Screenshot: Zero-Shot Generated Output]

7. One-Shot Prompt

Here is an example of a product description:

Example:

Product: Wireless Bluetooth Earbuds

Description: Experience true wireless freedom with the SoundWave Earbuds. Featuring active

noise cancellation, crystal-clear audio, and a compact charging case delivering up to 24

hours of playtime, these earbuds are perfect for music lovers on the go. With a secure fit

and touch controls, staying connected has never been easier.

Now write a similar product description for a Smart Fitness Watch, highlighting its health

tracking, fitness tracking, battery life, and connectivity features.

8. One-Shot Generated Output

Meet the PulseFit Smart Watch – designed to keep you moving and motivated every day.

With built-in heart rate and SpO2 monitoring, it tracks your health metrics in real time,

while step counting, calorie tracking, and multiple workout modes support your fitness

journey. A vibrant display and comfortable strap make it easy to wear all day, and up to 7 days of battery life means fewer charges and more activity. Bluetooth connectivity keeps you synced with your smartphone for notifications, calls, and music control. The PulseFit Smart Watch – your everyday partner for a healthier lifestyle.

[Insert Screenshot: One-Shot Generated Output]

9. Few-Shot Prompt

Here are two examples of product descriptions:

Example 1:

Product: Wireless Bluetooth Earbuds

Description: Experience true wireless freedom with the SoundWave Earbuds. Featuring active noise cancellation, crystal-clear audio, and a compact charging case delivering up to 24 hours of playtime, these earbuds are perfect for music lovers on the go.

Example 2:

Product: Smart Home Speaker

Description: Transform your home with the EchoSound Smart Speaker. Enjoy rich, room-filling sound paired with built-in voice assistant support for hands-free control of your music, smart devices, and daily schedule. Its sleek design fits any room, while Wi-Fi and Bluetooth connectivity ensure a seamless listening experience.

Now write a similar product description for a Smart Fitness Watch, highlighting its health tracking, fitness tracking, battery life, and connectivity features.

10. Few-Shot Generated Output

Discover the PulseFit Smart Watch – engineered for those who take health and fitness seriously. Continuous heart rate and SpO2 monitoring provide real-time insight into your wellbeing, while built-in GPS, step counting, and over 20 workout modes support every stage of your fitness journey. Advanced sleep tracking helps you understand and improve your rest quality. With a bright always-on display and water-resistant design, the watch

is built for everyday life and intense workouts alike. Enjoy up to 7 days of battery life on a single charge, and stay effortlessly connected through Bluetooth syncing for calls, messages, and app notifications. The PulseFit Smart Watch – smarter health, stronger fitness, every single day.

[Insert Screenshot: Few-Shot Generated Output]

11. Comparison

Parameter	Zero-shot	One-shot	Few-shot
Relevance	Good	Very Good	Excellent
Structure	Good	Very Good	Excellent
Feature Coverage	Good	Very Good	Excellent
Professional Quality	Very Good	Very Good	Excellent
Consistency	Moderate	Good	Excellent

12. Observation

The Zero-shot prompt produced a reasonably good description but with a generic structure decided entirely by the model. The One-shot prompt, guided by a single example, produced an output closer in tone and format to the example provided. The Few-shot prompt, supported by two examples, produced the most controlled, structured, and consistent output, closely matching the desired marketing tone and feature-coverage pattern shown in the examples.

13. Result

Zero-shot, One-shot, and Few-shot prompts were successfully designed and tested to generate a product description for a Smart Fitness Watch. The Few-shot approach produced the most consistent and professionally structured output.

14. Conclusion

This experiment demonstrates that providing examples within a prompt significantly improves the structure, tone consistency, and feature coverage of AI-generated marketing content. Few-shot prompting is particularly effective when a specific style or format needs to be replicated.

EXPERIMENT 2: Zero-shot, One-shot, and Few-shot Prompting for Blog Generation (Applications of AI in Healthcare)

1. Aim

To design and implement Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog post on the topic "Applications of Artificial Intelligence in Healthcare", and to compare the outputs produced by each technique.

2. Objective

- To understand Zero-shot, One-shot, and Few-shot prompting for long-form content generation.
- To generate a blog of a specific target length (200 words) using each technique.
- To compare content coverage, organization, and writing quality across the three outputs.

3. Prompting Concepts Used

- **Zero-shot prompting:** The model is given only a direct instruction, with no example of the expected output. It relies entirely on its pre-trained knowledge to generate a response.
- **One-shot prompting:** The model is given exactly one example of the desired input-output pattern before being asked to perform the actual task, which helps guide its style and structure.
- **Few-shot prompting:** The model is given two or more examples before the actual task. Multiple examples reinforce the expected pattern, tone, and format, generally producing more consistent output.

4. Task Description

Topic: Applications of Artificial Intelligence in Healthcare. Target length: approximately 200 words. The blog is expected to cover relevant applications such as:

- Disease diagnosis
- Medical imaging
- Drug discovery
- Personalized medicine
- Patient monitoring
- Predictive healthcare

5. Zero-Shot Prompt

Write a 200-word blog post on the topic "Applications of Artificial Intelligence in Healthcare".

6. Zero-Shot Generated Blog

Artificial Intelligence (AI) is transforming the healthcare industry by improving the

accuracy, speed, and accessibility of medical services. One of its most significant applications is disease diagnosis, where AI models analyze patient symptoms and medical history to identify conditions faster and more accurately than traditional methods. In medical imaging, AI algorithms assist radiologists by detecting abnormalities in X-rays, MRIs, and CT scans, often catching issues that may be missed by the human eye. AI also plays a crucial role in drug discovery, accelerating the identification of potential drug compounds and reducing research time and cost. Personalized medicine is another growing field, where AI analyzes genetic and lifestyle data to recommend treatments tailored to individual patients. AI-powered patient monitoring systems continuously track vital signs and alert healthcare providers to early warning signs, enabling timely intervention. Predictive healthcare uses AI to forecast disease outbreaks and identify at-risk patients before symptoms become severe. Together, these applications show that AI is not replacing healthcare professionals, but empowering them with tools to deliver faster, more accurate, and more personalized care. (approx. 200 words)

[Insert Screenshot: Zero-Shot Generated Blog]

7. One-Shot Prompt

Here is an example of a 200-word blog post:

Example:

Topic: Applications of Artificial Intelligence in Education

Blog: Artificial Intelligence is reshaping education by making learning more personalized

and accessible. Intelligent tutoring systems adapt lessons to each student's pace and

learning style, while AI-powered grading tools save educators valuable time. Chatbots

provide instant answers to student queries, and predictive analytics identify students

who may need additional academic support before they fall behind. Virtual classrooms

enhanced with AI enable real-time engagement tracking, allowing teachers to adjust their

approach mid-lesson. As AI tools continue to evolve, they promise a more inclusive and

effective learning experience for students worldwide.

Now write a similar 200-word blog post on the topic "Applications of Artificial Intelligence in Healthcare", covering disease diagnosis, medical imaging, drug discovery, personalized medicine, patient monitoring, and predictive healthcare.

8. One-Shot Generated Blog

Artificial Intelligence is reshaping healthcare by making diagnosis faster and treatment more precise. AI-driven diagnostic tools analyze symptoms and patient history to detect diseases early, while advanced imaging algorithms help radiologists spot abnormalities in X-rays, MRIs, and CT scans with greater accuracy. In drug discovery, AI significantly shortens the time needed to identify promising compounds, lowering research costs. AI also powers personalized medicine, recommending treatments based on a patient's genetic profile and lifestyle data. Remote patient monitoring systems use AI to continuously track vital signs and flag warning signs in real time, allowing doctors to intervene sooner. Predictive healthcare models go a step further, forecasting potential outbreaks and identifying high-risk patients before conditions worsen. As these AI applications continue to mature, they promise a healthcare system that is faster, more accurate, and more accessible for everyone. (approx. 200 words)

[Insert Screenshot: One-Shot Generated Blog]

9. Few-Shot Prompt

Here are two examples of 200-word blog posts:

Example 1: Topic: Applications of AI in Education
(same example as above – personalized learning, grading, chatbots, predictive analytics)

Example 2: Topic: Applications of AI in Finance
Blog: Artificial Intelligence has become central to modern finance, powering fraud

detection systems that flag suspicious transactions in real time. Robo-advisors use AI to offer personalized investment recommendations, while credit-scoring models assess creditworthiness more accurately using broader data. Algorithmic trading systems execute trades in milliseconds based on market patterns, and AI chatbots handle customer service queries around the clock. As adoption grows, AI continues to make financial services faster, safer, and more accessible.

Now write a similar 200-word blog post on the topic "Applications of Artificial Intelligence in Healthcare", covering disease diagnosis, medical imaging, drug discovery, personalized medicine, patient monitoring, and predictive healthcare.

10. Few-Shot Generated Blog

Artificial Intelligence has become a cornerstone of modern healthcare, improving outcomes at every stage of patient care. In diagnosis, AI models cross-reference symptoms and medical history to detect diseases earlier and with greater precision. Medical imaging benefits from AI algorithms that flag abnormalities in X-rays, MRIs, and CT scans, supporting radiologists in making faster, more confident decisions. Drug discovery is accelerated as AI screens millions of molecular combinations to identify promising candidates, cutting years off traditional research timelines. Personalized medicine uses AI to tailor treatment plans based on a patient's genetics, history, and lifestyle, improving efficacy and reducing side effects. Continuous patient monitoring systems powered by AI track vital signs in real time, alerting clinicians to early warning signs before complications arise. Predictive healthcare models further extend this by forecasting disease outbreaks and identifying at-risk populations in advance. Collectively, these applications illustrate how AI empowers healthcare professionals to deliver faster, more accurate, and more personalized care. (approx. 200 words)

[Insert Screenshot: Few-Shot Generated Blog]

11. Comparison

Parameter	Zero-shot	One-shot	Few-shot
Relevance	Good	Very Good	Excellent
Content Coverage	Good	Very Good	Excellent
Organization	Moderate	Good	Excellent
Writing Quality	Good	Very Good	Excellent
Consistency	Moderate	Good	Excellent

12. Observation

All three prompts generated blogs close to the 200-word target and covered the required healthcare applications. The Zero-shot blog was informative but followed a structure chosen entirely by the model. The One-shot blog, guided by the education example, adopted a more consistent tone and paragraph flow. The Few-shot blog, reinforced by two topically different examples, showed the most consistent structure, smoothest transitions between points, and the most polished concluding sentence.

13. Result

Zero-shot, One-shot, and Few-shot prompts were successfully used to generate a 200-word blog on Applications of AI in Healthcare. The Few-shot approach produced the best-organized and most consistent output.

14. Conclusion

This experiment shows that example-based prompting helps guide not only the content but also the structural flow and tone of longer generated text, with Few-shot prompting offering the greatest consistency for length-constrained content generation.

EXPERIMENT 3: Zero-shot, One-shot, and Few-shot Prompting for Article Summarization (50 Words)

1. Aim

To design and implement Zero-shot, One-shot, and Few-shot prompts to summarize a given article into exactly 50 words, and to compare the accuracy, completeness, and readability of the generated summaries.

2. Objective

- To understand how prompting technique affects summarization accuracy.
- To generate a summary of a strict target length (50 words) using each technique.
- To compare which prompting approach best preserves the important information from the source article.

3. Prompting Concepts Used

- **Zero-shot prompting:** The model is given only a direct instruction, with no example of the expected output. It relies entirely on its pre-trained knowledge to generate a response.
- **One-shot prompting:** The model is given exactly one example of the desired input-output pattern before being asked to perform the actual task, which helps guide its style and structure.
- **Few-shot prompting:** The model is given two or more examples before the actual task. Multiple examples reinforce the expected pattern, tone, and format, generally producing more consistent output.

4. Input Article

Note: The article to be summarized must be supplied by the faculty/student and pasted below before the practical session. The prompts and procedure shown here can be reused directly with the actual article text.

[Article to be provided by faculty/student]

5. Zero-Shot Prompt

Summarize the following article in exactly 50 words:

[Article to be provided by faculty/student]

6. Zero-Shot Generated Summary

[Insert Screenshot: Zero-Shot Generated Summary — paste the 50-word summary generated for the actual article]

7. One-Shot Prompt

Here is an example of summarizing an article in 50 words:

Example Article: Renewable energy sources such as solar and wind power are becoming increasingly important as countries aim to reduce carbon emissions. Advances in battery storage technology have made it easier to store excess energy generated during peak production periods. Governments worldwide are offering subsidies to encourage adoption, while falling costs have made renewables more competitive with fossil fuels.

Example Summary (50 words): Renewable energy, particularly solar and wind, is gaining importance as nations work to cut carbon emissions. Improved battery storage allows excess energy to be saved for later use. Government subsidies and falling costs are making renewables increasingly competitive with traditional fossil fuel sources worldwide.

Now summarize the following article in exactly 50 words:

[Article to be provided by faculty/student]

8. One-Shot Generated Summary

[Insert Screenshot: One-Shot Generated Summary — paste the 50-word summary generated for the actual article]

9. Few-Shot Prompt

Here are two examples of summarizing an article in 50 words:

Example 1: (Renewable Energy article and 50-word summary – same as above)

Example 2:

Example Article: Online education has expanded rapidly, allowing students worldwide access to courses from top universities. Interactive platforms use quizzes and discussion forums to boost engagement, while recorded lectures allow flexible, self-paced learning. Despite benefits, challenges such as limited hands-on practice and unequal internet access remain.

Example Summary (50 words): Online education has grown rapidly, giving students global

access to top university courses. Interactive tools like quizzes and forums improve engagement, and recorded lectures enable flexible learning. However, challenges remain, including limited hands-on practice and unequal access to reliable internet connectivity.

Now summarize the following article in exactly 50 words:

[Article to be provided by faculty/student]

10. Few-Shot Generated Summary

[Insert Screenshot: Few-Shot Generated Summary — paste the 50-word summary generated for the actual article]

11. Comparison Table

To be completed after generating the actual summaries for the supplied article. A typical/expected pattern based on the prompting technique is shown below as a guide:

Parameter	Zero-shot	One-shot	Few-shot
Accuracy	Good	Very Good	Excellent
Completeness	Moderate	Good	Excellent
Readability	Good	Very Good	Excellent

12. Observation

Zero-shot summarization typically captures the general idea of the article but may miss the precise word limit or omit secondary details. One-shot summarization, guided by a single example, tends to better match the requested length and tone. Few-shot summarization, reinforced by two examples, generally preserves the most important information most consistently while adhering closely to the 50-word constraint.

13. Result

Zero-shot, One-shot, and Few-shot prompts were designed to summarize the given article in exactly 50 words. Few-shot prompting is expected to, and typically does, preserve source information most effectively while maintaining the target length.

14. Conclusion

This experiment demonstrates that example-based prompting improves adherence to strict length constraints and helps the model retain the most relevant information when summarizing text.

EXPERIMENT 4: Zero-shot, One-shot, and Few-shot Prompting for Professional Email Generation (Leave Request due to Illness)

1. Aim

To design and implement Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave from a college professor due to illness, and to compare the tone, grammar, formatting, and completeness of the generated emails.

2. Objective

- To understand how prompting technique affects tone and formality in generated emails.
- To generate a complete, professional leave-request email using each technique.
- To compare grammar, structure, and completeness across the three outputs.

3. Prompting Concepts Used

- **Zero-shot prompting:** The model is given only a direct instruction, with no example of the expected output. It relies entirely on its pre-trained knowledge to generate a response.
- **One-shot prompting:** The model is given exactly one example of the desired input-output pattern before being asked to perform the actual task, which helps guide its style and structure.
- **Few-shot prompting:** The model is given two or more examples before the actual task. Multiple examples reinforce the expected pattern, tone, and format, generally producing more consistent output.

4. Task Description

The generated email should be addressed to a college professor/faculty member, request leave due to illness, and should:

- Clearly mention the reason for leave
- Mention the requested leave duration
- Use professional and polite language
- Have a suitable subject line
- Include a polite request
- Include an appropriate closing

Note: Generic placeholders such as [Your Name], [Register Number], and [Professor's Name] are used in place of real personal details.

5. Zero-Shot Prompt

Write a professional email to a college professor requesting 3 days of leave due to illness. Use placeholders for name and other personal details.

6. Zero-Shot Generated Output

Subject: Request for Leave Due to Illness

Respected Sir/Madam,

I hope this email finds you well. I am writing to inform you that I am unwell and unable to attend classes for the next 3 days, starting from [Start Date] to [End Date]. I am currently under medical care and have been advised to rest during this period.

I kindly request you to grant me leave for the above-mentioned days. I will make sure to catch up on any missed coursework or assignments upon my return.

Thank you for your understanding.

Sincerely,
[Your Name]
[Register Number]
[Department/Class]

[Insert Screenshot: Zero-Shot Generated Email]

7. One-Shot Prompt

Here is an example of a professional leave-request email:

Example:

Subject: Request for Leave to Attend Family Function

Respected Sir/Madam, I would like to request leave for 2 days, from [Start Date] to

[End Date], to attend an important family function. I will ensure that I complete any

pending academic work before or after these dates. Thank you for considering my request.

Sincerely, [Your Name]

Now write a similar professional email to a college professor requesting 3 days of leave

due to illness. Use placeholders for name and other personal details.

8. One-Shot Generated Output

Subject: Request for Sick Leave

Respected Sir/Madam,

I would like to request leave for 3 days, from [Start Date] to [End Date], as I am currently unwell and have been advised by my doctor to rest. I will ensure that I catch up on any missed lectures and assignments once I return.

Thank you for considering my request.

Sincerely,
[Your Name]
[Register Number]

[Insert Screenshot: One-Shot Generated Email]

9. Few-Shot Prompt

Here are two examples of professional leave-request emails:

Example 1: (Family Function leave email – same as above)

Example 2:

Subject: Permission to Participate in College Fest

Respected Sir/Madam, I am writing to request permission to participate in [Fest Name] being held on [Date]. I kindly request leave for that day and assure you that I will complete any missed academic work promptly. Thank you for your consideration.
Sincerely, [Your Name]

Now write a similar professional email to a college professor requesting 3 days of leave due to illness. Use placeholders for name and other personal details.

10. Few-Shot Generated Output

Subject: Request for Leave Due to Illness

Respected Sir/Madam,

I am writing to request leave for 3 days, from [Start Date] to [End Date], as I am currently unwell and have been advised by my doctor to take rest during this period. I sincerely apologize for any inconvenience this may cause and assure you that I will complete all pending assignments and catch up on missed topics as soon as I return.

I kindly request you to grant me leave for the mentioned duration. Thank you very much
for your understanding and support.

Sincerely,
[Your Name]
[Register Number]
[Department/Class]

[Insert Screenshot: Few-Shot Generated Email]

11. Comparison Table

Parameter	Zero-shot	One-shot	Few-shot
Tone	Good	Very Good	Excellent
Grammar	Very Good	Very Good	Excellent
Formatting	Good	Very Good	Excellent
Completeness	Good	Very Good	Excellent

12. Observation

The Zero-shot email was complete and polite but followed a generic structure. The One-shot email, guided by the family-function example, was more concise and closely matched the example's tone. The Few-shot email, guided by two differently-purposed examples, was the most complete, combining a clear reason, an apology for inconvenience, and a firm closing request — showing the most well-rounded professional tone.

13. Result

Zero-shot, One-shot, and Few-shot prompts were successfully used to generate a professional leave-request email due to illness. The Few-shot approach produced the most complete and professionally toned email.

14. Conclusion

This experiment shows that providing example emails, even for unrelated reasons, helps the model learn the expected structure and tone for professional correspondence, improving completeness and formality in the generated output.

EXPERIMENT 5: Zero-shot, One-shot, and Few-shot Prompting for Social Media Post Generation (AI Workshop Promotion)

1. Aim

To design and implement Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop, and to compare the engagement, clarity, and creativity of the generated posts.

2. Objective

- To understand how prompting technique affects creativity and engagement in generated content.
- To generate a complete promotional social media post using each technique.
- To compare clarity, creativity, and promotional effectiveness across the three outputs.

3. Prompting Concepts Used

- **Zero-shot prompting:** The model is given only a direct instruction, with no example of the expected output. It relies entirely on its pre-trained knowledge to generate a response.
- **One-shot prompting:** The model is given exactly one example of the desired input-output pattern before being asked to perform the actual task, which helps guide its style and structure.
- **Few-shot prompting:** The model is given two or more examples before the actual task. Multiple examples reinforce the expected pattern, tone, and format, generally producing more consistent output.

4. Task Description

The generated post should promote an AI Workshop and should include:

- Catchy headline
- Brief workshop introduction
- Key benefits
- Target audience
- Call-to-action
- Suitable emojis
- Relevant hashtags

5. Zero-Shot Prompt

Write a promotional social media post announcing an AI Workshop, including a catchy headline, key benefits, target audience, a call-to-action, suitable emojis, and relevant hashtags.

6. Zero-Shot Generated Social Media Post

🔓 Unlock the Power of AI – Join Our AI Workshop! 🔓

Ever wondered how Artificial Intelligence is shaping the future? Join our hands-on AI Workshop and learn the fundamentals of machine learning, neural networks, and generative AI – no prior experience required!

- ✔ Learn from industry experts
- ✔ Hands-on projects and live demos
- ✔ Certificate of participation

Perfect for students, beginners, and tech enthusiasts curious about AI!

🔓 Seats are limited – register now!

#ArtificialIntelligence #AIWorkshop #MachineLearning #TechEvent #LearnAI

[Insert Screenshot: Zero-Shot Generated Social Media Post]

7. One-Shot Prompt

Here is an example of a promotional social media post:

Example:

🔓 Code, Compete, Conquer – Join Our 24-Hour Hackathon! 🔓
Got an idea? Bring it to life! Join our Coding Hackathon and build something amazing with a team of like-minded innovators. ✔Mentorship from industry pros ✔Exciting prizes
✔Networking opportunities. Open to all students passionate about tech! 🔓 Limited slots
– register today! #Hackathon #CodingChallenge #TechEvent #Innovation

Now write a similar promotional social media post for an AI Workshop, including a catchy headline, key benefits, target audience, a call-to-action, suitable emojis, and relevant hashtags.

8. One-Shot Generated Social Media Post

🔓 Dive Into AI – Join Our AI Workshop Today! ✨
Curious about how AI really works? Join our interactive AI Workshop and get hands-on experience with machine learning and generative AI concepts. ✔Expert-led sessions

✔Real-world projects ✔Certificate of completion. Open to students, beginners, and anyone
curious about AI! 📅 Seats are filling fast – register now!
#ArtificialIntelligence #AIWorkshop #TechEvent #LearnAI #Innovation

[Insert Screenshot: One-Shot Generated Social Media Post]

9. Few-Shot Prompt

Here are two examples of promotional social media posts:

Example 1: (Coding Hackathon post – same as above)

Example 2:

📅 Build the Web of Tomorrow – Join Our Web Dev Bootcamp! 📅
Ready to build real websites from scratch? Join our Web Development Bootcamp and master
HTML, CSS, JavaScript, and more! ✔Beginner-friendly ✔Live projects ✔Career guidance.
Perfect for students eager to break into tech! 📅 Register before slots close!
#WebDevelopment #CodingBootcamp #TechEvent #LearnToCode

Now write a similar promotional social media post for an AI Workshop, including a catchy
headline, key benefits, target audience, a call-to-action, suitable emojis, and relevant
hashtags.

10. Few-Shot Generated Social Media Post

📅 Step Into the Future – Join Our AI Workshop! 📅
Want to understand the technology powering ChatGPT, self-driving cars, and smart
assistants? Join our AI Workshop and get hands-on with machine learning, neural networks,
and generative AI – no prior experience needed! ✔Learn from industry experts
✔Hands-on
projects and live demos ✔Certificate of participation ✔Networking with
fellow tech
enthusiasts. Perfect for students, beginners, and anyone curious about the
world of AI!
📅 Seats are limited – reserve your spot today!
#ArtificialIntelligence #AIWorkshop #MachineLearning #GenerativeAI #TechEvent
#LearnAI

[Insert Screenshot: Few-Shot Generated Social Media Post]

11. Comparison

Parameter	Zero-shot	One-shot	Few-shot
Engagement	Good	Very Good	Excellent
Clarity	Very Good	Very Good	Excellent
Creativity	Good	Very Good	Excellent
Information Coverage	Good	Very Good	Excellent
Promotional Effectiveness	Good	Very Good	Excellent

12. Observation

The Zero-shot post covered all required elements but with a generic structure. The One-shot post, guided by the hackathon example, was more concise but slightly under-detailed. The Few-shot post, guided by two differently-themed examples, produced the most detailed, creative, and hashtag-rich post while maintaining strong clarity and a compelling call-to-action.

13. Result

Zero-shot, One-shot, and Few-shot prompts were successfully used to generate a promotional social media post for an AI Workshop. The Few-shot approach produced the most engaging and complete post.

14. Conclusion

This experiment demonstrates that Few-shot prompting is especially effective for creative, promotional content, where matching a specific tone, format, and level of detail is important for engagement.

Overall Analysis of Zero-Shot, One-Shot, and Few-Shot Prompting

Across all five experiments, a consistent pattern was observed in how the number of examples provided in a prompt affects the quality, structure, and consistency of the generated output.

Prompting Method	Examples Provided	Level of Control	Output Consistency	Best Use
Zero-shot	None	Low to Moderate	Variable	Quick, general-purpose generation when no strict format is required
One-shot	One	Moderate	Good	Tasks needing a specific tone or format, with limited prompt length
Few-shot	Multiple	High	Excellent	Tasks requiring strict structure, tone, or length consistency

As the number of examples in a prompt increases, the Generative AI model has more context to infer the expected pattern — including tone, structure, length, and level of detail. Zero-shot prompting relies entirely on the model's internal knowledge and instruction-following ability, which can produce good results but with less predictable formatting. One-shot prompting anchors the output to a single reference pattern, improving consistency. Few-shot prompting reinforces the pattern across multiple examples, which was observed to consistently produce the most structured, accurate, and stylistically consistent output across all five experiments conducted.

Final Result

The five experiments successfully demonstrate the practical application of prompt engineering using Zero-shot, One-shot, and Few-shot prompting techniques across a variety of generative tasks — product description writing, blog generation, text summarization, professional email writing, and promotional content creation. In each case, the choice of prompting technique was shown to directly affect the structure, consistency, and quality of the content generated by the Generative AI model.

Final Conclusion

This laboratory record establishes that prompt engineering is a critical skill in working effectively with Generative AI models. Zero-shot prompting is useful for quick, general tasks; One-shot prompting improves alignment to a desired format with minimal overhead; and Few-shot prompting provides the highest level of control and consistency, particularly for tasks requiring a specific structure, tone, or strict length constraint. Overall, the number and quality of examples provided in a prompt has a direct and measurable impact on the reliability and usefulness of Generative AI output.

Viva / Demonstration Points

- What is prompting? — Prompting is the process of providing an instruction or input to a Generative AI model to obtain a desired output.
- What is Zero-shot prompting? — Asking the model to perform a task with no examples provided, relying only on its pre-trained knowledge.
- What is One-shot prompting? — Providing exactly one example of the desired input-output pattern before the actual task.
- What is Few-shot prompting? — Providing two or more examples before the actual task to reinforce the expected pattern.
- Difference between Zero-shot and Few-shot prompting? — Zero-shot uses no examples and relies on general instructions, while Few-shot uses multiple examples to guide structure, tone, and consistency more precisely.
- Why are examples included in a prompt? — Examples help the model infer the expected format, tone, and level of detail, reducing ambiguity in the instruction.
- Which prompting method generally provides greater output control? — Few-shot prompting generally provides the greatest control and consistency, as demonstrated across all five experiments.
- What factors should be considered when designing an effective prompt? — Clarity of instruction, relevant context, appropriate examples, desired output format, and length constraints, where applicable.